
Neutrinos via Charm Decays in Astrophysical Sources

TALK IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR THE DEGREE OF BACHELOR OF SCIENCE

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Particle Physics

- Point A
- Point B
 - Part 1
 - Part 2
- Point C
- Point D



Multimessenger Astronomy

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Astrophysical Sources

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Cross Sections

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Spectral Distributions

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Computation

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Results

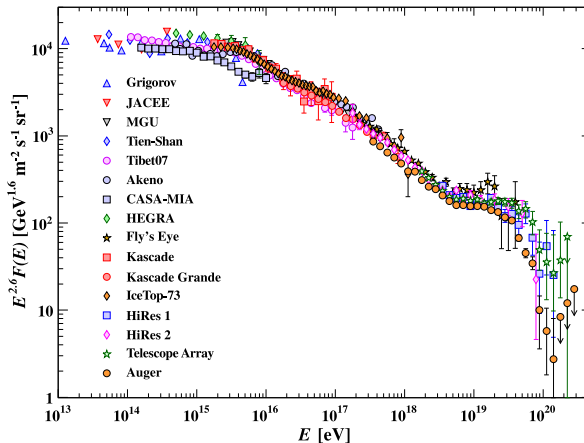
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Conclusion & Outlook

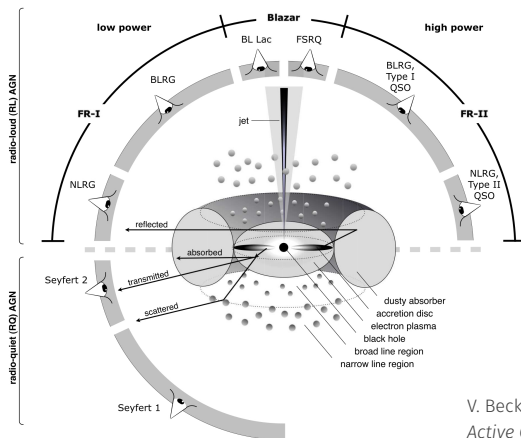
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All-Particle CR Spectrum & Cutoff



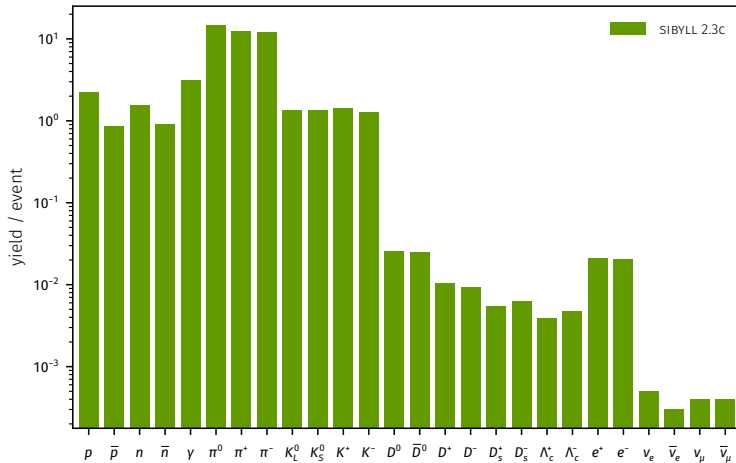
K. A. Olive et. al.
Chin. Phys. C (2014)

Unified AGN Classification Model



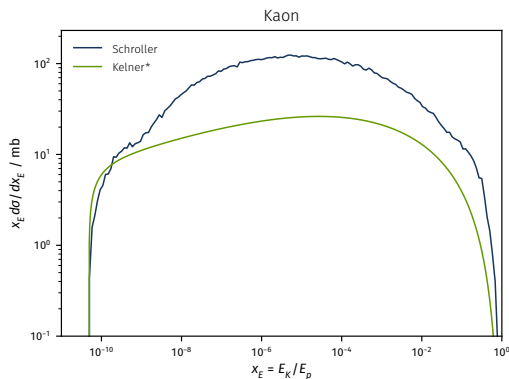
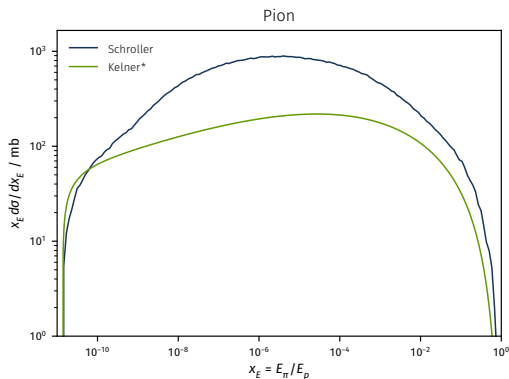
V. Beckmann, C. Shrader
Active Galactic Nuclei (2012)

Event Generation



Data Courtesy
M. Schroller

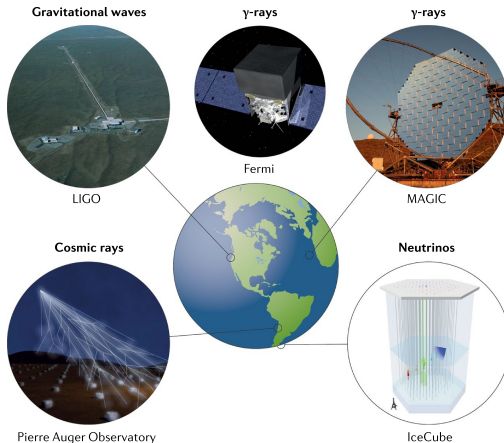
Weighted Energy Differential Cross Section



Data Courtesy M. Schrollier (SIBYLL 2.3C)

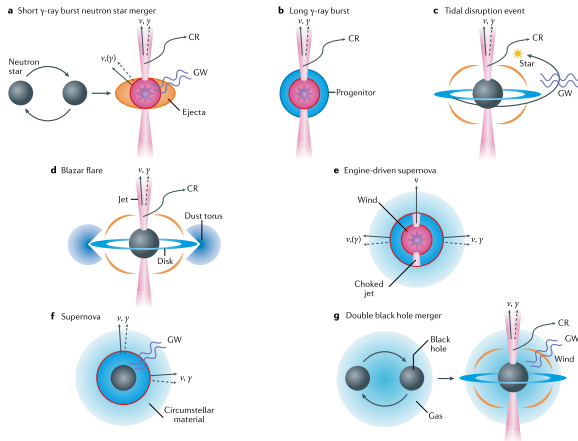
S. R. Kelner et. al. *Phys. Rev. D* (2006)

Messenger Detection



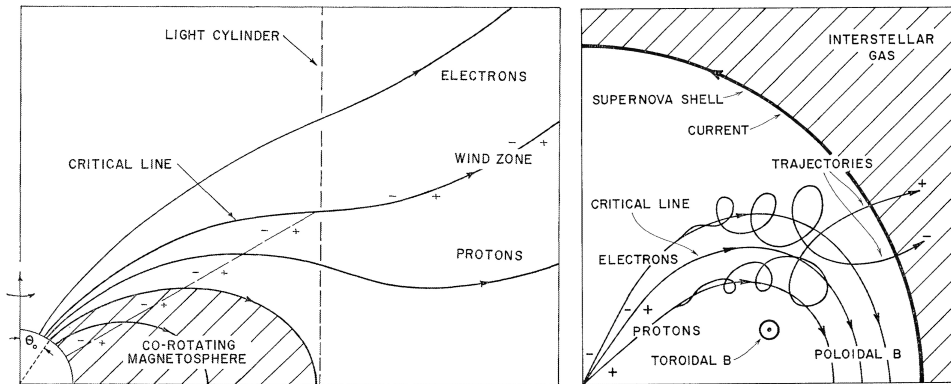
P. Mészáros et. al.
Nat. Rev. Phys. (2019)

Multimessenger Sources



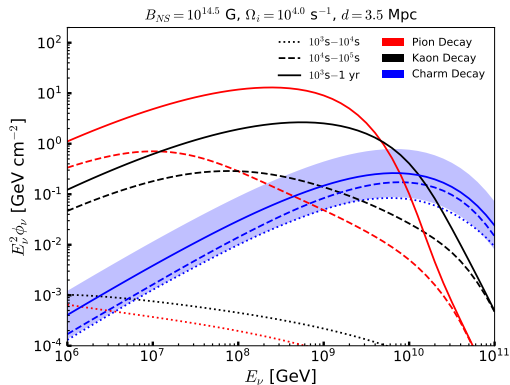
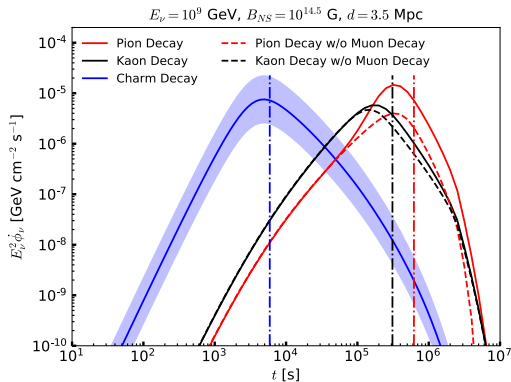
P. Mészáros et. al.
Nat. Rev. Phys. (2019)

Pulsar Particle Acceleration



P. Goldreich, W. H. Julian *ApJ* (1969)

Magnetar Neutrino Flux & Fluence Comparison



J. A. Carpio et. al.
Phys. Rev. D (2020)